

Indian Institute of Space Science and Technology
Thiruvananthapuram



Course on Python Programming, Data Processing & Visualization

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Python Programming Practice

Input · Output · Variables · Operators · Expressions

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Question 1: Personal Information

Problem

Write a Python program to accept the following information from the user:

- Name
- Age
- Place

Display all the information entered by the user.

Hint

- Use `input()` to read the values.
- Store each value in a separate variable.
- Use `print()` to display the information.

Answer 1: Personal Information

Question: Accept name, age and place and display them.

```
name = input("Enter your name: ")
age = input("Enter your age: ")
place = input("Enter your place: ")

print("Name:", name)
print("Age:", age)
print("Place:", place)
```

Concepts: `input()`, variables, `print()`

Question 2: Student Information

Problem

Write a Python program to accept:

- Student name
- Roll number
- Course
- College name

Display the details as a student profile.

Hint

Create one variable for each piece of information and use `input()` to accept the values.

Answer 2: Student Information

Question: Accept student details and display them.

```
name = input("Enter name: ")
roll_no = input("Enter roll number: ")
course = input("Enter course: ")
college = input("Enter college: ")

print("\n--- Student Information ---")
print("Name:", name)
print("Roll Number:", roll_no)
print("Course:", course)
print("College:", college)
```

Concepts: Variables, strings, `input()`

Question 3: Annual Salary

Problem

Write a Python program that accepts an employee's monthly salary and calculates the annual salary.

$$\text{Annual Salary} = \text{Monthly Salary} \times 12$$

Hint

Use `float(input())` to accept the salary and multiply it by 12.

Answer 3: Annual Salary

Question: Calculate annual salary from monthly salary.

```
monthly_salary = float(
    input("Enter monthly salary: ")
)

annual_salary = monthly_salary * 12

print("Annual salary =", annual_salary)
```

Concepts: `float()`, multiplication, variables

Question 4: Rectangle Area

Problem

Write a Python program to accept the length and breadth of a rectangle and calculate its area.

$$\text{Area} = \text{Length} \times \text{Breadth}$$

Hint

Take length and breadth as inputs and multiply them.

Answer 4: Rectangle Area

Question: Calculate the area of a rectangle.

```
length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))

area = length * breadth

print("Area =", area)
```

Concepts: Numeric input, arithmetic operators

Question 5: Basic Calculator

Problem

Write a Python program that accepts two numbers and calculates:

- Addition
- Subtraction
- Multiplication
- Division

Hint

Use the arithmetic operators:

$$+, \quad -, \quad *, \quad /$$

Answer 5: Basic Calculator

Question: Perform four basic arithmetic operations.

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

addition = a + b
subtraction = a - b
multiplication = a * b
division = a / b

print("Addition =", addition)
print("Subtraction =", subtraction)
print("Multiplication =", multiplication)
print("Division =", division)
```

Concepts: Arithmetic operators, variables

Question 6: Celsius to Fahrenheit

Problem

Write a Python program to convert temperature from Celsius to Fahrenheit.

$$F = \frac{9}{5}C + 32$$

Hint

- Accept Celsius temperature as a number.
- Apply the given formula.
- Display the Fahrenheit value.

Answer 6: Celsius to Fahrenheit

Question: Convert Celsius temperature to Fahrenheit.

```
celsius = float(
    input("Enter temperature in Celsius: ")
)

fahrenheit = (celsius * 9 / 5) + 32

print("Temperature in Fahrenheit =",
      fahrenheit)
```

Concepts: Formula-based calculation

Question 7: Kilometre Conversion

Problem

Write a Python program to accept a distance in kilometres and convert it into:

- Metres
- Centimetres

Use:

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ km} = 100000 \text{ cm}$$

Hint

Use multiplication for both conversions.

Answer 7: Kilometre Conversion

Question: Convert kilometres to metres and centimetres.

```
km = float(input("Enter distance in km: "))

metres = km * 1000
centimetres = km * 100000

print("Metres =", metres)
print("Centimetres =", centimetres)
```

Concepts: Unit conversion, multiplication

Question 8: Time Conversion

Problem

Write a Python program to accept a number of seconds and convert it into:

- Hours
- Minutes
- Remaining seconds

For example:

$$3665 \text{ seconds} = 1 \text{ hour } 1 \text{ minute } 5 \text{ seconds}$$

Hint

Use:

- `//` for integer division
- `%` to find the remainder

Answer 8: Time Conversion

Question: Convert total seconds into hours, minutes, and seconds.

```
seconds = int(input("Enter seconds: "))

hours = seconds // 3600
remaining = seconds % 3600

minutes = remaining // 60
seconds = remaining % 60

print("Hours =", hours)
print("Minutes =", minutes)
print("Seconds =", seconds)
```

Concepts: //, %, integer division

Question 9: Simple Interest

Problem

Write a Python program to calculate simple interest.

$$SI = \frac{P \times R \times T}{100}$$

where:

- P = Principal
- R = Rate of interest
- T = Time in years

Hint

Accept all three values using `float(input())`.

Answer 9: Simple Interest

Question: Calculate simple interest using the formula.

```
p = float(input("Enter principal: "))
r = float(input("Enter rate: "))
t = float(input("Enter time in years: "))

si = (p * r * t) / 100

print("Simple Interest =", si)
```

Concepts: Formula, arithmetic operations

Question 10: Salary with Allowance

Problem

An employee has a basic monthly salary and receives a monthly allowance.
Write a program to calculate the total monthly salary.

$$\text{Total Salary} = \text{Basic Salary} + \text{Allowance}$$

Hint

Accept both values and add them.

Answer 10: Salary with Allowance

Question: Calculate total monthly salary.

```
basic_salary = float(
    input("Enter basic salary: ")
)

allowance = float(
    input("Enter monthly allowance: ")
)

total_salary = basic_salary + allowance

print("Total monthly salary =",
      total_salary)
```

Concepts: Variables, addition

Question 11: Salary with HRA and DA

Problem

Calculate an employee's gross salary using:

- HRA = 20% of basic salary
- DA = 10% of basic salary

Use:

$$\text{Gross Salary} = \text{Basic} + \text{HRA} + \text{DA}$$

Hint

First calculate HRA and DA separately and then add them to the basic salary.

Answer 11: Salary with HRA and DA

Question: Calculate gross salary using HRA and DA.

```
basic = float(input("Enter basic salary: "))

hra = basic * 20 / 100
da = basic * 10 / 100

gross_salary = basic + hra + da

print("Basic Salary =", basic)
print("HRA =", hra)
print("DA =", da)
print("Gross Salary =", gross_salary)
```

Concepts: Percentages, arithmetic operations

Question 12: Salary Deduction

Problem

An employee has a gross salary. A deduction of 10% is made from the salary.
Calculate:

- Deduction amount
- Net salary

$$\text{Net Salary} = \text{Gross Salary} - \text{Deduction}$$

Hint

First calculate 10% of the gross salary.
Then subtract the deduction from the gross salary.

Answer 12: Salary Deduction

Question: Calculate net salary after a 10% deduction.

```
gross_salary = float(
    input("Enter gross salary: ")
)

deduction = gross_salary * 10 / 100
net_salary = gross_salary - deduction

print("Deduction =", deduction)
print("Net Salary =", net_salary)
```

Concepts: Percentage, subtraction

Question 13: Salary Increment

Problem

An employee receives a salary increment of 10%.
Write a program to calculate:

- Increment amount
- New salary

$$\text{Increment} = \frac{\text{Salary} \times 10}{100}$$

$$\text{New Salary} = \text{Salary} + \text{Increment}$$

Hint

Calculate the increment first and then add it to the current salary.

Answer 13: Salary Increment

Question: Calculate the new salary after a 10% increment.

```
salary = float(input("Enter current salary: "))

increment = salary * 10 / 100
new_salary = salary + increment

print("Increment =", increment)
print("New Salary =", new_salary)
```

Concepts: Percentage, addition

Question 14: Salary Based on Range

Problem

An employee receives an increment based on their current salary:

- Below 30000: 10% increment
- 30000–60000: 7% increment
- Above 60000: 5% increment

Calculate the new salary.

Hint

Use:

- `if`
- `elif`
- `else`

Answer 14: Salary Based on Range

Question: Calculate salary increment based on salary range.

```
salary = float(input("Enter salary: "))

if salary < 30000:
    increment = salary * 10 / 100

elif salary <= 60000:
    increment = salary * 7 / 100

else:
    increment = salary * 5 / 100

new_salary = salary + increment

print("Increment =", increment)
print("New Salary =", new_salary)
```

Concepts: `if-elif-else`, percentage

Question 15: Employee Salary Slip

Problem

Write a Python program to generate an employee salary slip.
Accept:

- Employee name
- Basic salary

Calculate:

- HRA = 20% of basic salary
- DA = 10% of basic salary
- PF = 12% of basic salary
- Gross salary
- Net salary

$$\text{Gross} = \text{Basic} + \text{HRA} + \text{DA}$$

$$\text{Net} = \text{Gross} - \text{PF}$$

Hint

Calculate each component separately and then combine them.

Answer 15: Employee Salary Slip

Question: Generate an employee salary slip.

```
name = input("Enter employee name: ")
basic = float(input("Enter basic salary: "))

hra = basic * 20 / 100
da = basic * 10 / 100
pf = basic * 12 / 100

gross = basic + hra + da
net = gross - pf

print("\n===== SALARY SLIP =====")
print("Employee Name :", name)
print("Basic Salary  :", basic)
print("HRA           :", hra)
print("DA            :", da)
print("Gross Salary  :", gross)
print("PF            :", pf)
print("Net Salary    :", net)
```

Concepts: Input, variables, percentages, arithmetic

Question 16: Augmented Assignment Operators

Problem

Write a Python program to demonstrate augmented assignment operators by starting with a number and applying:

- `+=` (add and assign)
- `-=` (subtract and assign)
- `*=` (multiply and assign)
- `/=` (divide and assign)

Hint

`x += 5` is a shorthand for `x = x + 5`. The same pattern applies to `-=`, `*=`, and `/=`.

Answer 16: Augmented Assignment Operators

Question: Demonstrate augmented assignment operators.

```
x = 10
print("Initial value:", x)

x += 5
print("After x += 5:", x)

x -= 3
print("After x -= 3:", x)

x *= 2
print("After x *= 2:", x)

x /= 4
print("After x /= 4:", x)
```

Concepts: +=, -=, *=, /=

Question 17: Comparing Two Numbers

Problem

Write a Python program to accept two numbers and check:

- Are they equal?
- Is the first greater than the second?
- Is the first less than or equal to the second?

Hint

Use the comparison operators ==, >, and <=. Each comparison gives a `True` or `False` result.

Answer 17: Comparing Two Numbers

Question: Compare two numbers using comparison operators.

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("Equal?", a == b)
print("First greater?", a > b)
print("First <= Second?", a <= b)
```

Concepts: ==, >, <=, comparison operators

Question 18: Eligibility Check

Problem

A person is eligible to vote only if:

- Age is 18 or above, **and**
- They hold a valid ID (yes/no)

Write a Python program to accept age and ID status, then display whether the person is eligible to vote.

Hint

Use `and` to combine both conditions. Use `input()` for the ID status and compare it to `"yes"`.

Answer 18: Eligibility Check

Question: Check voting eligibility using logical operators.

```
age = int(input("Enter age: "))
has_id = input("Do you have a valid ID (yes/no): ")

eligible = age >= 18 and has_id == "yes"

print("Eligible to vote?", eligible)
```

Concepts: and, comparison operators, boolean result

Question 19: Expression Evaluation

Problem

Write a Python program that stores three numbers and evaluates the following expression:

$$\text{result} = a + b \times c$$

Then evaluate it again using parentheses:

$$\text{result} = (a + b) \times c$$

Display both results and observe the difference.

Hint

Python follows operator precedence: multiplication happens before addition, unless parentheses say otherwise.

Answer 19: Expression Evaluation

Question: Evaluate expressions with and without parentheses.

```
a = 10
b = 5
c = 2

result1 = a + b * c
result2 = (a + b) * c

print("a + b * c =", result1)
print("(a + b) * c =", result2)
```

Concepts: Operator precedence, parentheses, expressions

Question 20: Roots of a Quadratic Equation

Problem

Write a Python program to accept the coefficients a, b, c of a quadratic equation

$$ax^2 + bx + c = 0$$

and calculate its roots. Assume the roots are real.

Hint

Use the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Use `** 0.5` for the square root.

Answer 20: Roots of a Quadratic Equation

Question: Calculate the roots of a quadratic equation.

```
a = float(input("Enter a: "))
b = float(input("Enter b: "))
c = float(input("Enter c: "))

discriminant = b ** 2 - 4 * a * c

root1 = (-b + discriminant ** 0.5) / (2 * a)
root2 = (-b - discriminant ** 0.5) / (2 * a)

print("Root 1 =", root1)
print("Root 2 =", root2)
```

Concepts: Input, type conversion, **, arithmetic expressions

Thank You!